

he can obtain about 20 bp more yield by investing in synthetic EUR Treasuries rather than Bunds in the 1Y sector and about 45 bp in the 5Y segment.

- Within US Treasuries, there are several obvious outliers. Some of these are due to their benchmark status and repo specialness, and their spread could be adjusted by the technique of external explanatory variables. Others represent genuine richness and cheapness, i.e. candidates for trades. At the short end of the yield curve, micro bond switches between USTs and Bunds seem practicable, as the pricing of USTs and Bunds versus each other in that sector supports the convergence.

In the third step, one can now apply all the techniques derived in Chapter 8 also to relative value analysis between bonds issued in different currencies. In particular, one can add external explanatory variables to the optimization. For example, based on the results of Figure 17.1, it seems reasonable to include short maturity USTs, Gilts, and Bunds in the set *Bonds* and to calculate *global* external explanatory variables, such as a *global* benchmark premium. Incorporating this result into the global asset allocation decisions between short maturity USTs, Gilts, and Bunds, it is possible to select the cheapest and richest bonds while taking into account the usual richness of benchmarks, as explained in Chapter 8.

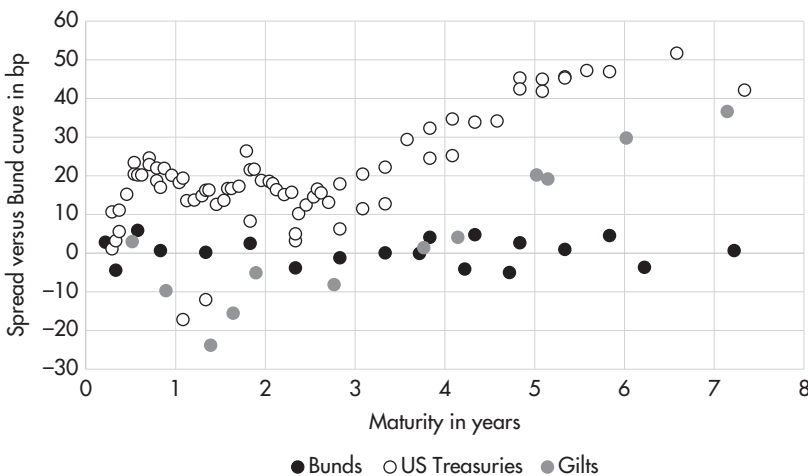


FIGURE 17.1 Richness and cheapness of global bonds swapped into EUR versus the fitted Bund yield curve.

Source: Authors.

Rich/Cheap Analysis through SOFR Spreads for Bonds Denominated in Different Currencies

Based on the arguments above, one can consider the par asset swap spreads of bonds denominated in or basis swapped into USD as alternative rich/cheap indicators (potentially after some adjustments). When using this alternative, step 1.3, i.e. the reverse asset swapping of floating (SOFR) into fixed payments, is skipped. Also the second step, i.e. the calculation of a fitted curve, is omitted and the spreads of the basis swapped bonds over USD SOFR are directly used as a global rich/cheap measure. In this case, the asset allocation decisions in step 3 are based on the (adjusted) SOFR asset swap spreads of basis swapped bonds.

Whereas the fitted curve (step 2) is not needed, the regression approach from Chapter 8 can be transferred to obtain external explanatory variables. In other words, when replacing the fitted curve with SOFR spreads as a rich/cheap measure, from our innovation to obtain the fitted curve and the external explanatory variables in one single optimization, only the regression versus external explanatory variables is left. One can then run a multiple regression of the SOFR spreads versus a number of external explanatory variables, such as benchmark status, and thereby obtain a *global* estimate for the benchmark premium also in this alternative. Moreover, the desired adjustment of bond yields for specialness effects described above can be incorporated into this regression framework.

Figure 17.2 depicts the perspective on global bond RV using the second alternative, also as of October 17, 2023.

Several observations can be explained by the theory presented in the previous (and next) chapters:

- The overall upward-sloping curves are in line with the upward-sloping CDS curves and haircut schedules, as discussed in Chapter 12. And the exceptionally steep haircut schedule of the BoE (Table 18.2) contributes to Gilts having the steepest SOFR–ASW curve.
- The ordering of 5Y Bunds < Gilts < Treasuries from rich to cheap is at least qualitatively in line with the levels of the 5Y CDS on the issuing governments. Quantitatively, however, the actual difference between SOFR spreads exceeds in longer maturity segments the difference predicted by the adjusted CDS levels. A portion of the remaining difference can be explained by the different haircuts. As concluded in Chapter 12 (Figure 12.4), incorporating haircuts is thus also an important element for modeling the relationships between global swap spreads.
- In comparison with the historical situation depicted in Figure 16.1, US Treasuries and Bunds seem to have changed places. This can also be

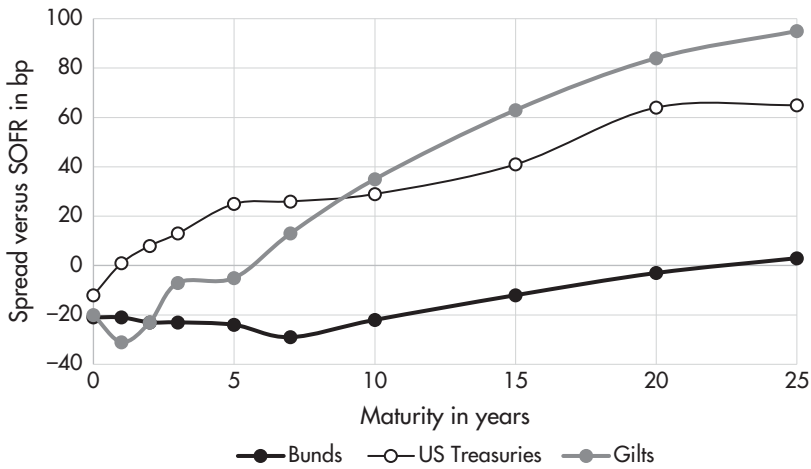


FIGURE 17.2 SOFR-ASW curves of global (basis swapped) bonds.

Source: Authors.

explained by the increasing CDS level of the US (which hardly traded at all in 2012 and if it did at very low levels) relative to Germany over the last decade. Despite the switch, the connection via investment arbitrage between both remains strong and hence the consequences for the equilibrium discussed in Chapter 16 still hold true.

- While the models explain a certain “fair” richness for Bunds due to the relatively low CDS level of Germany and the haircut schedules, in longer maturity segments, Bunds appear excessively rich. This points to additional driving factors not captured in the models, such as supply/demand imbalances. If these are deemed to be ephemeral, one could position for the excessive richness of Bunds to revert to the “fair” richness of Bunds as justified by the CDS, etc. This could be done by selling Bunds versus buying Treasuries or Gilts, both versus SOFR. The selection of specific bonds is supported by the rich/cheap numbers from Figure 17.1, or alternatively by using the SOFR spreads of the individual bonds directly.

Chapter 18 will deal with these remaining influencing factors, some of which are unobservable, such as the shadow costs of the marginal market participant. As a consequence, understanding and predicting the SOFR spreads observed in Figure 17.2 form a goal which allows only partial and approximative solutions.

Other Factors Affecting Swap Spreads

INTRODUCTION

In previous chapters, we discussed the relative valuations of swaps and bonds by applying no-arbitrage pricing considerations under the simplifying assumptions that there are no material market frictions and that market participants operate without any material binding constraints. In this chapter, we consider the effects of relaxing these assumptions. We start by considering the effects of haircuts applied to bonds and the related margins applied to swaps. We then discuss the effects of regulatory constraints that have been introduced in various jurisdictions. With these constraints in mind, we then turn our attention to the effects that the changing supply of bonds can have on relative valuations and close the chapter by discussing the implications that these issues present for relative value analysis of bonds and swaps.

HAIRCUTS AND MARGINS

The term *haircut* refers to the difference between the market value of the collateral posted in a repo transaction and the amount of cash that the lender is willing to lend against this collateral. For example, if I post US Treasuries with a market value of USD 100 million as collateral in a repo transaction, my repo counterparty might be willing to lend me USD 95 million against this collateral. If I were buying these bonds, I'd need to obtain the USD 5 million difference with some other source of funds. And the cost of these additional funds needs to be included in any relative value analysis involving the bonds. For example, if my source for these additional funds was my reserve balance at the Federal Reserve, the cost of these additional funds would be the Fed's Interest on Reserve Balances (IORB) rate, currently (October 2023) 5.40%, fairly close to the current value of SOFR, 5.30%. On the other hand, if the source of additional funds was equity capital, the cost of the additional funds would be significantly higher. For reference, the biggest banks in the US recently have been reporting costs of equity in the region of 10%.

So the effect of a haircut depends on the size of the haircut and on the cost of the funds that are used to replace the haircut.

Haircuts vary as a function of the maturity and credit quality of the bonds posted as collateral. For example, if one posts EUR 100 million of 5Y German Bunds as collateral in a repo transaction with the ECB, the ECB will currently lend EUR 98.5 million against that collateral. But if one posts EUR 100 million of 30Y Bunds as collateral, the ECB will lend only EUR 95 million. And if one posts 30Y Italian BTPs as collateral, the ECB will lend only EUR 86.5 million.

The ostensible logic in this arrangement is that the value of the collateral is constantly changing, so that the market value of the bonds serving as collateral may not be sufficient to cover the costs in the event the borrower of cash were to default on the loan in the future. And the thinking is that the prices of longer-term bonds tend to be more volatile than the prices of shorter-term bonds and that the prices of bonds issued by lower-rated issuers tend to be more volatile than the prices of bonds issued by higher-rated issuers. So to protect against the greater risk when lending against bonds with more volatile prices, repo counterparties tend to arrange larger haircuts for these bonds.

Repo haircuts in the private market also vary by the characteristics of the counterparties. For example, a recent note from the Federal Reserve¹ contains an estimate that 74% of US Treasury repo borrowing by hedge funds occurs with no haircut at all or even with a negative haircut.² At the same time, roughly 20% of hedge fund borrowing of Treasuries involve haircuts in excess of 2%. So we see that these arrangements differ considerably across counterparties.

Central banks are key participants in the repo market, particularly for banks. For example, the Federal Reserve has both a Repo Facility in which it lends cash against eligible collateral and a Reverse Repo Facility in which it accepts cash in exchange for bonds. The current (October 2023) haircut schedule for US bills, notes, bonds, TIPS, and STRIPS is shown in Table 18.1.

The corresponding table for UK Gilts submitted as collateral at the Bank of England is shown in Table 18.2.

Note that the haircut schedule of the BoE is much steeper than that of the Fed. An immediate implication is that, *ceteris paribus*, the Gilt yield curve swapped into dollars should be steeper than the US Treasury yield curve. And this was precisely our observation in Figure 17.2. To be clear, this assumes that the excess costs (above the respective risk-free rate) of financing the respective haircuts are positive and identical. It also assumes that the credit risk of the

¹Banegas and Phillip (2023).

²A negative haircut involves the value of cash lent being *greater* than the market value of the bonds posted as collateral.

TABLE 18.1 Federal Reserve Haircut Schedule

Maturity	Bills (%)	Notes and bonds (%)	TIPS (%)	STRIPS (%)
Less than 10Y	1.0	2.0	2.9	3.8
10Y–20Y	–	2.9	5.7	6.5
20Y–30Y	–	2.9	5.7	9.9

TABLE 18.2 Bank of England Haircut Schedule

Maturity	Coupon Bonds (%)	Zero Coupon Bonds (%)
Floating	0.5	0.5
Less than 1Y	0.5	0.5
1Y–3Y	1.5	1.5
3Y–5Y	2.0	2.0
5Y–10Y	3.0	3.5
10Y–20Y	5.5	6.5
20Y–30Y	7.0	9.5
More than 30Y	8.5	15.0

two issues is the same. These assumptions are too restrictive to be practical in this case, but they do help to illustrate one of the general effects of haircuts on relative value analysis of bonds.

The haircut schedule at the ECB is more complicated, as haircuts for sovereign debt are tiered by issuer as well as by maturity. Table 18.3 illustrates the haircuts currently in effect for some of the larger sovereign issuers.

Note the significant difference between the haircuts applied to bonds issued by Germany, France, and Spain and the haircuts applied to bonds issued by Italy.

One way to assess the effect of the differences in haircuts is to assume a value for the excess cost of financing the haircut (i.e. the cost of funds in excess of the repo rate) and to calculate the effect on the yield of each bond, by maturity. Figure 18.1 shows the results of this exercise for zero-coupon bonds, assuming the excess cost of funds for the haircut is 600 basis points.³

Note that the effect of haircuts on the spread between BTPs and Bunds is about 47 bp for 10Y and 54 bp for 30Y zero-coupon yields.

³Reported costs of equity for large European banks in Europe recently have been in the vicinity of 10%, while ESTR is currently 3.9%.

TABLE 18.3 ECB Haircut Schedule

Maturity	Germany, France, Spain: zero coupon (%)			
	Germany, France, Spain: bullet (%)	Italy: bullet (%)	Italy: zero coupon (%)	
Less than 1Y	0.5	0.5	5.0	5.0
1Y to 3Y	1.0	2.0	6.0	7.0
3Y to 5Y	1.5	2.5	8.5	10.0
5Y to 7Y	2.0	3.0	10.0	11.5
7Y to 10Y	3.0	4.0	11.5	13.0
10Y to 15Y	4.0	5.0	12.5	14.0
15Y to 30Y	5.0	6.0	13.5	15.0
More than 30Y	6.0	9.0	14.0	17.0

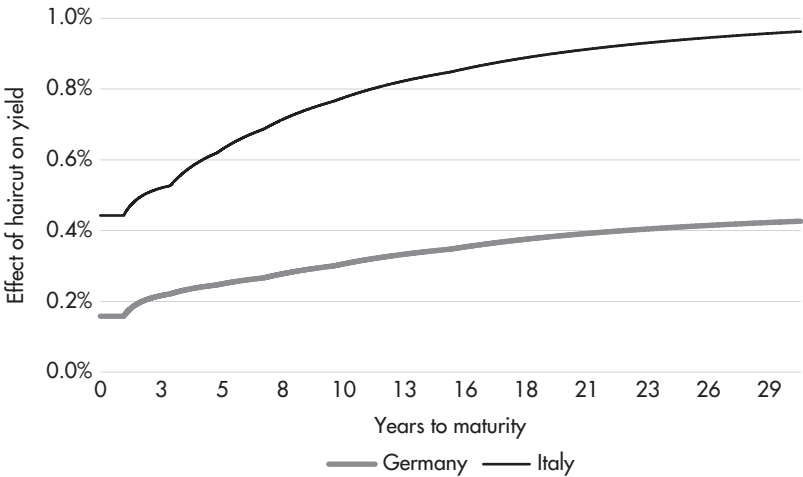


FIGURE 18.1 Effect of ECB haircut schedule on German and Italian zero-coupon bond yields.
Source: Authors.

Another consideration to note is the possibility of haircut arbitrage. For example, under most market conditions, it may not make sense to post zero-coupon bonds as collateral to the Bank of England, particularly if it's possible to reconstitute those zero-coupon bonds into a low-coupon issue, as the Bank of England will often lend more for a bullet issue than for a set of zero-coupon bonds that replicates the cash flows of the bullet.

Margin and Collateral Agreements

Just as leveraged bond positions typically involve haircuts, derivative instruments, like interest rate swaps, credit default swaps, and basis swaps, typically involve collateral to be posted as margin, which limits the risk of a swap counterparty in the event the other counterparty to the swap were to default.

Rules governing the margining of derivatives can be bilateral in the case of two parties that agree to clear a trade bilaterally. But they can also be standardized, as is the case with central clearinghouses of the sort used by futures exchanges. But even in the case of futures exchanges, the margin terms offered to end users may differ based on the futures commission merchant (FCM) that facilitates the trade, as FCMs are able to specify many of the terms that govern the treatment of collateral posted as margin.

For example, a typical hedge fund does business with multiple dealers and is likely to have negotiated different collateral agreements with these dealers. So a hedge fund that wants to execute a cross-currency basis swap may find that it gets slightly better pricing from one dealer but that this advantage is more than offset by the relatively advantageous collateral terms offered by another dealer.

The complexities that can arise when collateralizing derivative trades require traders and analysts to be familiar with the terms of these collateral agreements when assessing the relative valuations of various instruments. In practice, these considerations are often ignored, as the effects of these agreements are often fairly modest. But there have been plenty of instances in which unsuspecting traders have been picked off by counterparties who take advantage of the terms of collateral agreements, and we encourage traders to be familiar with the terms of the collateral agreements negotiated with their counterparties.

REGULATORY CONSIDERATIONS

Prior to the financial crisis of 2008–2009, many banks operated with relatively few constraints on their balance sheets. If a bank was interested in undertaking a new investment, it could compare the expected rate of return on the project to the marginal cost of the funds required to pursue the project. If the expected rate of return exceeded the cost of funds by an amount sufficient to justify any additional risks presented by the project, the bank would often proceed with the investment. The bank's balance sheet would increase, but this typically presented no problem.

Subsequent to the financial crisis, regulators in most of the developed world introduced a considerable number of additional regulations that posed

material, often binding constraints on bank balance sheets. Many of these fell under the rubric of the Basel III capital accords. A great deal has been written about the Basel III accords, and we won't attempt a detailed discussion here. But we will outline some of the key characteristics for traders, portfolio managers, and analysts conducting relative value analysis of the fixed income markets. In particular, we'll comment on capital requirements, leverage ratios, funding requirements, and market risk regulations.

Box 18.1

Shadow Costs

A bank optimizing the return on its balance sheet will have to deal with the concept of shadow costs if its balance sheet is subject to constraints. In the field of mathematical optimization, a shadow cost is the amount by which the function being optimized could improve in the event that a binding constraint was relaxed. In the context of a bank with a constrained balance sheet evaluating a new investment, the shadow cost is the amount by which return on capital could be increased by relaxing the constraint.

As a simplified example, consider a brand new bank, with cash on the asset side of its balance sheet and with customer deposits and contributed equity on the liability side of the balance sheet.

Assume that this bank pays an interest rate of 2% per year on new deposits and faces no constraints on its balance sheet. We might expect the bank to fund any and all new investments available, as long as the expected return on those investments is greater than its 2% cost of funds (with some suitable adjustment for risk). If we ranked the expected (risk-adjusted) returns of these investments in descending order, we'd expect to find that the lowest return on capital was close to 2%, since the bank would pursue all projects with expected returns greater than 2% and would decline all projects with expected returns less than 2%.

Now let's consider the same bank in the presence of balance sheet constraints. For example, let's imagine that the bank first funds a project with an expected return of 10% and that it subsequently funds projects with expected returns of 9%, 8%, and 7%. But let's also assume that after funding the project with the expected return of 7%, the bank finds that its CET1 capital ratio has declined to 4.5%. At this point, it faces two choices if it wants to pursue another investment.

First, it could issue more equity. For example, if it raised 10% of the funds required for the project by selling equity and 90% by accepting more

customer deposits, its rate of return hurdle for the new project would be the blended cost of capital, which depends not only on the amount paid to customer deposits but also on the bank's cost of equity. If the cost of equity was 10%, this blended cost of funds would be 2.8% – i.e. 80 bp greater than the marginal cost of funds when the balance sheet was unconstrained.

Even if the bank decided to raise only 4.5% of the required funds by issuing equity, thereby keeping the CET1 capital ratio at 4.5%, it would face a blended cost of capital of 2.36% – i.e. 36 bp greater than its cost of funds in the event its balance sheet was unconstrained.

A second alternative would be for the bank to dispose of an existing asset to make room for the new investment. We presume the asset to be liquidated would be the asset with the lowest expected return, in this case the project with an expected return of 7%. In this case, the marginal cost of the funds required to pursue the new investment would be 7%, regardless of the amount the bank was paying on customer deposits.

In the first approach, the binding constraint is taken to be the leverage ratio alone, and additional equity is issued in order to allow the bank to pursue the new investment. In this case, we can think of the shadow cost of this constraint to be 2.36% (assuming the CET1 ratio of 4.5% is maintained).

In the second approach, the bank faces two binding constraints: the capital ratio imposed by regulators; and its desire to avoid selling additional equity, which typically would be a self-imposed constraint. The shadow cost of the self-imposed constraint in this example is 7%.

In this example, the shadow costs of the two constraints were quite different. In practice, we'd expect most banks to structure their balance sheets so that the various shadow costs were more similar. And we should stress that optimization of bank balance sheets in practice is a very complicated process, with a considerable number of regulatory and self-imposed constraints, often in multiple jurisdictions, and with market costs and market opportunities changing constantly.

Capital Requirements

The Basel III process increased the amount of *Tier 1 Capital* that banks need to hold against risk-weighted assets (RWAs). Tier 1 capital is the sum of *Common Equity Tier 1* (CET1) *Capital* (which includes common equity, reserves, and retained earnings) and *Additional Tier 1* (AT1) *Capital* (which includes contingent convertible debt, and perpetual preferred stock). Risk-weighted assets are the weighted sum of bank assets (and certain off-balance sheet items), weighted by the perceived risks of each asset.